

CAPE CAMPBELL, New Zealand

WMO: 935970

Lat: 41.7264S

Lon: 174.2761E

Elev: 20

StdP: 14.69

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	41.6	42.8	27.6	21.6	45.0	30.8	25.0	45.9	41.6	50.6	38.9	50.8	15.4	200	0.822

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.5	68.7	60.8	67.2	60.7	66.0	60.3	63.1	66.3	62.2	65.4	61.5	64.6	24.2	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.7	82.2	64.9	60.8	79.5	64.2	59.9	77.2	63.7	28.7	66.6	28.0	65.6	27.5	64.8	67.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 39.2	35.2	32.5	38.9	73.8	1.4	2.4	37.9	75.6	37.1	77.0	36.4	78.3	35.4	80.1		
(5)			34.1	64.7	1.6	1.3	32.9	65.6	31.9	66.4	31.0	67.1	29.8	68.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	55.2	61.1	61.4	60.2	56.9	54.0	50.4	48.9	49.6	51.5	53.4	55.9	59.2	(6)
(7)		DBStd	5.30	2.89	2.58	2.96	3.00	3.23	3.12	2.88	2.85	3.09	3.12	3.22	3.07	(7)
(8)		HDD50	161	0	0	0	1	5	32	55	41	19	7	1	0	(8)
(9)		HDD65	3593	124	104	151	244	340	438	498	476	404	359	274	181	(9)
(10)		CDD50	2050	345	319	315	207	130	44	22	30	64	112	177	285	(10)
(11)		CDD65	9	4	3	1	0	0	0	0	0	0	0	0	1	(11)
(12)		CDH74	2	1	0	0	0	0	0	0	0	0	0	0	1	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	17.4	18.2	16.8	17.4	16.4	16.7	16.3	16.4	16.2	18.0	18.7	19.0	18.3	(14)
(15)	Precipitation	PrecAvg	25.7	2.0	1.6	1.8	2.1	2.3	2.9	2.4	2.1	2.0	2.0	1.8	(15)	
(16)		PrecMax	41.8	4.2	5.4	4.9	5.8	6.6	6.0	7.8	6.5	4.5	4.6	5.1	4.0	(16)
(17)		PrecMin	14.6	0.0	0.0	0.1	0.2	0.5	0.8	0.6	0.4	0.0	0.2	0.2	0.2	(17)
(18)		PrecStd	6.4	1.1	1.3	1.1	1.5	1.4	1.4	1.7	1.4	1.3	1.1	1.3	1.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.5	70.7	70.0	65.9	63.3	59.5	57.3	58.2	61.1	63.7	67.4	69.7	(19)
(20)			MCWB	61.0	61.7	59.9	58.1	56.0	53.2	51.6	51.0	53.0	54.4	56.1	58.4	(20)
(21)		2%	DB	68.4	68.2	67.2	63.7	61.3	57.7	55.7	55.9	58.6	61.0	63.6	66.7	(21)
(22)			MCWB	61.5	61.9	60.1	57.6	55.5	52.5	50.8	50.5	52.2	54.5	56.1	59.5	(22)
(23)		5%	DB	66.7	66.7	65.8	62.4	60.0	56.5	54.5	54.8	57.3	59.4	61.9	64.9	(23)
(24)			MCWB	61.0	61.4	60.0	57.2	54.9	51.9	50.1	50.1	51.9	53.9	55.7	58.9	(24)
(25)		10%	DB	65.4	65.6	64.6	61.2	58.9	55.3	53.4	53.8	56.0	58.1	60.5	63.6	(25)
(26)			MCWB	60.5	60.9	59.7	56.7	54.4	51.2	49.4	49.6	51.4	53.2	55.2	58.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.6	64.6	63.0	60.6	58.7	55.3	53.7	53.0	55.2	57.2	59.3	62.2	(27)
(28)			MCDWB	67.9	67.6	66.3	62.9	60.9	57.4	55.7	55.4	58.6	60.5	62.9	65.4	(28)
(29)		2%	WB	63.0	63.4	61.9	59.2	57.3	53.9	52.0	51.9	53.9	55.8	57.9	61.1	(29)
(30)			MCDWB	66.4	66.4	65.1	61.9	59.8	56.3	54.4	54.4	56.9	59.1	61.5	64.3	(30)
(31)		5%	WB	62.1	62.5	61.1	58.3	56.1	52.9	50.9	51.1	52.9	54.8	57.0	60.2	(31)
(32)			MCDWB	65.4	65.6	64.4	61.2	58.9	55.5	53.5	53.7	56.0	58.2	60.5	63.5	(32)
(33)		10%	WB	61.2	61.7	60.3	57.4	54.9	51.8	50.0	50.3	52.1	53.9	56.1	59.3	(33)
(34)			MCDWB	64.5	64.7	63.7	60.5	57.9	54.7	52.7	53.1	55.3	57.3	59.6	62.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.6	6.5	6.4	6.1	5.8	6.0	5.6	5.9	6.3	7.0	7.1	6.7	(35)
(36)			MCDBR	8.4	7.9	7.6	7.0	6.8	7.0	6.4	6.5	7.3	8.4	8.7	8.3	(36)
(37)			MCWBR	5.1	5.1	5.0	5.0	5.1	5.7	5.4	5.1	5.2	5.7	5.5	5.2	(37)
(38)		5% WB	MCDWB	7.0	6.9	6.3	5.9	5.3	5.5	5.1	5.3	6.3	6.9	7.3	6.8	(38)
(39)			MCWBR	4.7	4.8	4.7	4.8	4.6	5.1	4.8	4.7	5.0	5.3	5.0	4.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.341	0.335	0.319	0.311	0.291	0.288	0.287	0.300	0.320	0.328	0.333	0.335	(40)	
(41)		taud	2.574	2.607	2.647	2.647	2.680	2.646	2.652	2.634	2.555	2.555	2.559	2.602	(41)	
(42)		Ebn at Noon	313	307	298	279	265	254	263	279	292	305	313	317	(42)	
(43)		Edh at Noon	33	30	27	23	19	18	19	23	29	31	33	32	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2096	1860	1453	963	653	491	566	823	1204	1631	2009	2089	(44)	
(45)		RadStd	145	156	81	72	38	29	40	47	72	108	160	144	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	+0.63	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page